

Analysis of Residual Ionospheric Errors in GNSS-RO Bending Angles Using Ray Tracing

Jihyeok Park
Korea Advanced Institute of
Science and Technology
Daejeon, Republic of Korea
1999jimmy@kaist.ac.kr

Jaehee Chang
Korea Advanced Institute of
Science and Technology
Daejeon, Republic of Korea
chang.cathy7@kaist.ac.kr

Andrew K. Sun
Korea Advanced Institute of
Science and Technology
Daejeon, Republic of Korea
kiyoungsun@kaist.ac.kr

Jiyun Lee
Korea Advanced Institute of
Science and Technology
Daejeon, Republic of Korea
jiyunlee@kaist.ac.kr

Abstract—Global Navigation Satellite System Radio Occultation (GNSS-RO) is a satellite-based remote sensing technique that estimates neutral atmospheric parameters—such as temperature, humidity, and pressure—by analyzing the refraction and delay of GNSS signals received by Low Earth Orbit (LEO) satellites as they pass through the Earth’s atmosphere. Accurate retrieval of neutral atmospheric bending angles requires effective mitigation of ionospheric effects. While the conventional dual-frequency linear combination of L1 and L2 bending angles can remove most contributions of ionospheric delays, large-scale Residual Ionospheric Errors (RIEs), caused by differences in signal paths of L1 and L2 through the ionosphere, may still persist. This study presents a comprehensive analysis of RIEs across solar cycle, signal height, and various geophysical conditions to guide the development of more accurate correction model for enhanced atmospheric retrievals. To achieve this, a 3D observation-based ionospheric environment was constructed from COSMIC-2 electron density measurements using a 1D double-Chapman model, and ray tracing simulations were conducted to compute the RIEs. The analysis demonstrates that the predominantly negative RIE becomes more negative with increasing solar activity and signal altitude. Geographically, its magnitude is most pronounced during the daytime and at small solar zenith angles, with a high concentration of large errors occurring at the two crests of the equatorial ionization anomaly.

Keywords—GNSS Radio Occultation, Bending Angle, Residual Ionospheric Error, Ray Tracing Simulation

I. INTRODUCTION

Global Navigation Satellite System Radio Occultation (GNSS-RO) is a satellite-based technique for obtaining vertical profiles of atmospheric parameters like temperature, pressure, and humidity. These profiles are vital for improving numerical weather prediction and for climate monitoring [1, 2]. The technique relies on precise measurements of how much GNSS signals bend as they pass through the Earth’s atmosphere. However, the signals are bent by both the neutral atmosphere and the ionosphere. At higher altitudes, the ionosphere becomes the dominant source of error, and its effect must be removed to ensure accurate neutral atmosphere retrievals.

The major component of the ionospheric bending angle is inversely proportional to the square of the signal frequency. This property allows for the removal of most of the ionospheric effect by using a linear combination of the bending angles from two different frequencies, typically f_{L1} and f_{L2} [3]. The standard correction formula is:

$$\frac{f_{L1}^2 \alpha_{L1} - f_{L2}^2 \alpha_{L2}}{f_{L1}^2 - f_{L2}^2} = \alpha_{NA} + \alpha_{RIE} \quad (1)$$

α_{NA} is the neutral atmosphere bending angle and α_{RIE} denotes the Residual Ionospheric Error (RIE). Equation (1) is valid under the assumption that the rays follow identical geometric paths; in dispersive ionosphere, however, frequency-dependent refraction separates the L1/L2 ray trajectories, so the correction leaves a RIE spanning broad altitudes [4]. To improve the accuracy of GNSS-RO results, this study systematically analyzes how the RIE varies with solar activity levels, signal altitude, and key geophysical parameters using an observation-based model from COSMIC-2 data.

II. DATA PROCESSING & RESULTS

This research utilizes an observation-based approach to realistically model the ionospheric environment. Rather than relying solely on a theoretical electron density model, it fits a 1D double-Chapman model to actual electron

density measurements from COSMIC-2 ionospheric profiles provided by the COSMIC Data Analysis and Archive Center (CDAAC). This fitting process is crucial for converting discrete observational data into a continuous, analytic function suitable for subsequent ray tracing simulations. The resulting three-dimensional representation of the ionosphere accurately captures key features (E and F layer peak electron densities, peak heights, and scale heights).

Using this custom-built ionospheric model, the study performs ray tracing simulations with the Radio Occultation Simulator for Atmospheric Profiling (ROSAP) algorithm. This simulation estimates the signal path of GNSS signals as they propagate through the ionosphere, allowing for the precise calculation of ionospheric bending angles for both L1 and L2 frequencies [6]. These calculated bending angles are then used to derive the RIE, which is the remaining error after dual-frequency linear combination, for parametric analysis.

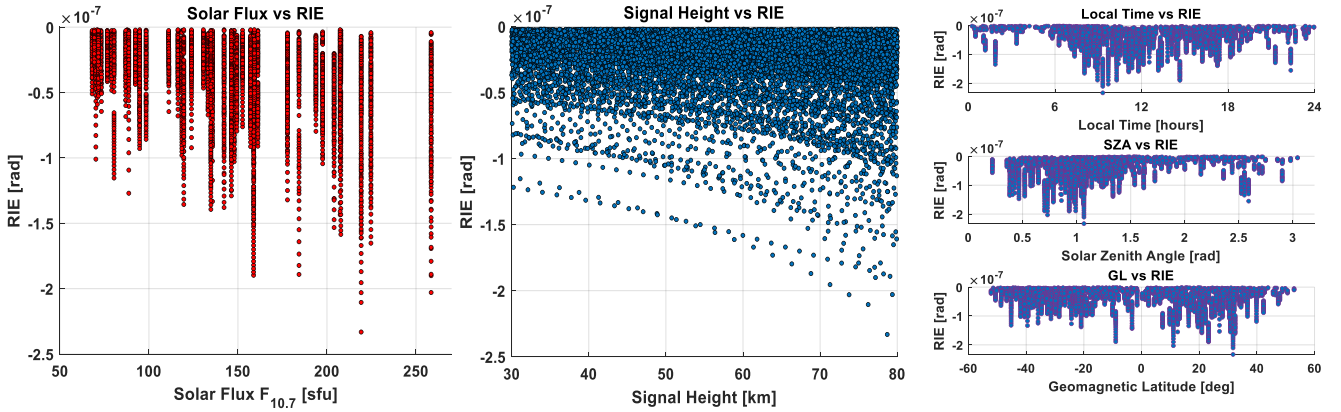


Figure 1 Scatter plots of GNSS-RO RIE versus (left) solar flux F10.7, (middle) signal height, and (right) local time, solar zenith angle, and geomagnetic latitude-grouped of illustrate geographic dependence

As shown in Figure 1, the predominantly negative RIE is strongly dependent on several factors. First, it becomes more strongly negative with increasing solar activity (F10.7) and signal height, with increased variability observed at higher factors. Second, geographic drivers further modulate the RIE; its greatest magnitudes are observed during the daytime and at small solar zenith angles. Furthermore, a pronounced lower envelope is evident, with the error reaching its large absolute values in two distinct crests near $|\text{Geomagnetic Latitude}| \approx 20^\circ$, where the equatorial ionization anomaly predominantly occurs.

III. SUMMARY

This work builds an ionospheric model from real electron-density measurements and performs 3D ray-tracing simulation to compute the RIE in GNSS-RO bending angles. The RIE increases with ionization intensity—stronger during high solar activity—and grows with signal height. It also shows clear geographic dependencies across local time, geomagnetic latitude, and solar zenith angle. These findings provide physics-based guidance for developing improved RIE correction schemes, ultimately enhancing the accuracy of high-altitude atmospheric retrievals.

Acknowledgement

This work was supported by the KASA, Korea Aero Space Administration, under Grant 2022M1A3C2069728, Future Space Education Center. The authors would like to thank Dr. Stig Syndergaard from DMI (Danish Meteorological Institute) for providing simulation software.

References

- [1] E. R. Kursinski, G. A. Hajj, J. T. Schofield, R. T. Linfield, and K. R. Hardy, "Observing Earth's atmosphere with radio occultation measurements using the Global Positioning System," *J. Geophys. Res.*, vol. 102, pp. 23429–23465, 1997.
- [2] G. A. Hajj, E. R. Kursinski, L. J. Romans, W. I. Bertiger, and S. S. Leroy, "A technical description of atmospheric sounding by GPS occultation," *J. Atmos. Solar-Terr. Phys.*, vol. 64, pp. 451–469, 2002.
- [3] V. V. Vorob'ev and T. G. Krasil'nikova, "Estimation of the accuracy of the atmospheric refractive index recovery from Doppler shift measurements at frequencies used in the NAVSTAR system," *Izv. Atmos. Ocean. Phys.*, vol. 29, pp. 602–609, 1994.
- [4] S. Syndergaard, "On the ionosphere calibration in GPS radio occultation measurements," *Radio Sci.*, vol. 35, pp. 865–883, 2000.
- [5] J. Danzer, B. Scherllin-Pirscher, and U. Foelsche, "Systematic residual ionospheric errors in radio occultation data and a potential way to minimize them," *Atmos. Meas. Tech.*, vol. 6, pp. 2169–2179, 2013.
- [6] P. Hoeg et al., "Derivation of atmospheric properties using a radio occultation technique," Danish Meteorological Institute, Technical Report DMI SR 95-4, Copenhagen, 1996.